

Instructions: Answer all questions. Show your work for each problem by following the four problem-solving steps.

Speed and Velocity Formulas

$$s = \frac{d}{t} \quad v = \frac{\Delta x}{t} \quad d = s t \quad \Delta x = v t$$

Reminders

- Speed is a scalar; velocity is a vector (include the direction).
- SI units of speed and velocity: m/s.
- If a vector has no given direction, use positive (+) or negative (-).
- Resultant velocity: add collinear vectors in the same direction, subtract opposite ones.

1. A roller skater travels 240 m along a straight path in 30 s. What is her speed?

1. Given / Finding:

2. Equation:

3. Plug in and solve:

4. Check units & direction:

2. A horse gallops at a constant speed of 12 m/s for 15 s. How far does it travel?

1. Given / Finding:

2. Equation:

3. Plug in and solve:

4. Check units & direction:

3. A drone flies 360 m north in 45 s, as shown. What is its velocity?



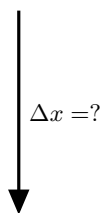
1. Given / Finding:

2. Equation:

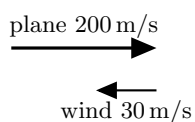
3. Plug in and solve:

4. Check units & direction:

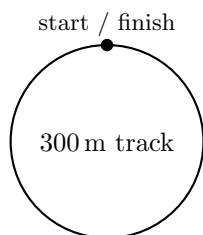
4. A submarine descends at a constant velocity of 4 m/s for 25 s, as shown. What is its displacement?



1. Given / Finding:
 2. Equation:
 3. Plug in and solve:
 4. Check units & direction:
5. A plane flies east at 200 m/s into a wind that blows 30 m/s west. What is the plane's resultant velocity? Use $v = v_1 + v_2$.



1. Given / Finding:
 2. Equation:
 3. Plug in and solve:
 4. Check units & direction:
- Exit Ticket** _____
6. A bicyclist rides around a circular track at a constant speed of 15 m/s. The track is 300 m around, as shown. After 60 s:



- (a) What is the total distance traveled?

- (b) What is the total displacement?